



MANIPAL SCHOOL OF INFORMATION SCIENCES

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Master of Engineering – M.E in Computer Science and Engineering

Course Plan

Course Name	:	Data Structures and Algorithms
Course Code	:	ESE 5102
Academic Year	:	2026-2027
Semester	:	I
Name of the Course Coordinator	:	Jyothsna V Shetty
Name of the Program Coordinator	:	Dr. Sathyendranath Malli

Date:	08-08-2026
Signature of Program Coordinator	
with Date	

Date:	07-08-2026
Signature of Course Coordinator	
with Date	



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Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering – M.E in Computer Science and Engineering, programme are as follows

PEO No.	Education Objective
PEO 1	Enable to draw upon fundamental and advanced knowledge to apply analytical and computational approaches to solve technological problems in computer science.
PEO 2	Introduce state of art technologies in computer science and inculcate ethical practices to make industry-ready professionals.
PEO 3	Promote scientific and societal advancement through research and entrepreneurship.



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Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering – M.E in Computer Science and Engineering, graduates will be able to:

PO 1	An ability to independently carry out research /investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	An ability to develop software based on theoretical principles and practical knowledge.
PO 5	An ability to demonstrate knowledge of the underlying principles and evaluation methods for analysing and decision-making.



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1. Course Plan

1.1 Primary Information

Course Name	:	Data Structures and Algorithms [ESE 5102]
L-T-P-C	:	3-0-0-3
Contact Hours	:	36 Hours
Pre-requisite	:	C Programming
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No.ofContact Hours	Program Outcomes (POs)	BL
CO1	Analyse the implementations of elementary data structures.	12	PO5	3
CO2	Design and implementation of data structures like trees and graphs.	15	PO4	4
CO3	Analyse the concepts of divide and conquer, dynamic programming, greedy, and back tracking techniques to efficiently solve problems.	9	PO3	4



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1.3 Assessment Plan

Components	Mid semester exam (MSE)	Flexible assessments	End semester exam (ESE) / Makeup exam
Duration	90 minutes	Assignments are to be submitted within one week of posting	180 minutes
Weightage	0.3	0.2	0.5
Typology of questions	Apply and Analyse	Apply and Analyse	Apply and Analyse
Pattern	Each question may 3/4/5/10 marks	Solve problems by designing the data structure and functionalities for the given set of problems	Answer all 10 questions of 10 marks each, each question may have sub questions of 3/4/5/6/7 marks
Schedule	As per academic calendar.	Assignment submission: October 2026	As per academic calendar.
Topics covered	Linked List, Stack, Queue, Circular Queue, Sorting, Trees	Time Complexity, Linked Lists, Stacks, Queues, Sorting, Trees	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.



1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Algorithms, Performance Analysis	C01
L2	Time complexity, Space complexity	C01
L3	Analysis of recursive function	C01
L4	Defining the data structure and code organisation	C01
L5	Linked List: Insert at beginning and end	C01
L6	Linked list: search, max min problem	C01
L7	Linked list: insert in between, intersection	C01
L8	Linked List: Deletions	C01
L9	Stack: array and linked list	C01
L10	Queue, circular queue: array and linked list	C01
L11	Sorting: Bubble, selection, insertion	C02



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L12	Sorting: Quick sort	C02
L13	Sorting: Merge sort	C02
L14	Basic Terminology – Implementation of Trees	C02
L15	Binary tree – properties Binary search tree	C02
L16	Binary tree – insertion, traversal	C02
L17	Binary search tree- deletion	C02
L18	Heap Property: Heap Sort	C02
L19	Searching: linear and binary	C02
L20	Hash table: closed hashing	C02
L21	Hash Table: open hashing	C02
L22	Hash table: problem solving	C02
L23	Graph: properties, adjacency matrix	C02
L24	Graph: Adjacency list	C02
L25	Graph: BFS, DFS	C02



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L26	Greedy algorithm: Minimum spanning tree - Prims algorithm, Kruskal's Algorithm	C03
L27	Single source shortest path: Dijktras algorithm	C03
L28	Greedy algorithm: Knapsack, optimal storage on tapes	C03
L29	Greedy algorithm : Job Scheduling	C03
L30	Divide and Conquer : Finding Max and Min	C03
L31	Back tracking: graph coloring, Nqueens problem	C03
L32	Back tracking: Sum of subset	C03
L33	Dynamic programing: Matrix chain multiplication	C03
L34	Dynamic programming: All pair shortest path	C03
L35	Revision	C01
L36	Revision	C01



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1.5 References

1. "Introduction to Algorithms" Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest.
2. "Data Structures & Algorithms" Aho, Hopcroft and Ulmann.
3. "Data structures and algorithm analysis in C" Mark Allen Weiss.
4. "Computer Algorithms" : Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran.
5. MOOC: [https://in.coursera.org/learn/algorithms-on-graphs?specialization=data-structures-](https://in.coursera.org/learn/algorithms-on-graphs?specialization=data-structures-algorithms#enroll)
6. [algorithms#enroll](https://in.coursera.org/learn/algorithms-on-graphs?specialization=data-structures-algorithms#enroll)

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.



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1.7 Course Timetable

I Semester						Room: LG01-LH08			
	8-9	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Monday				DSA					
Tuesday									
Wednesday				DSA					
Thursday									
Friday				DSA					
Saturday									



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1.8 Assessment Plan

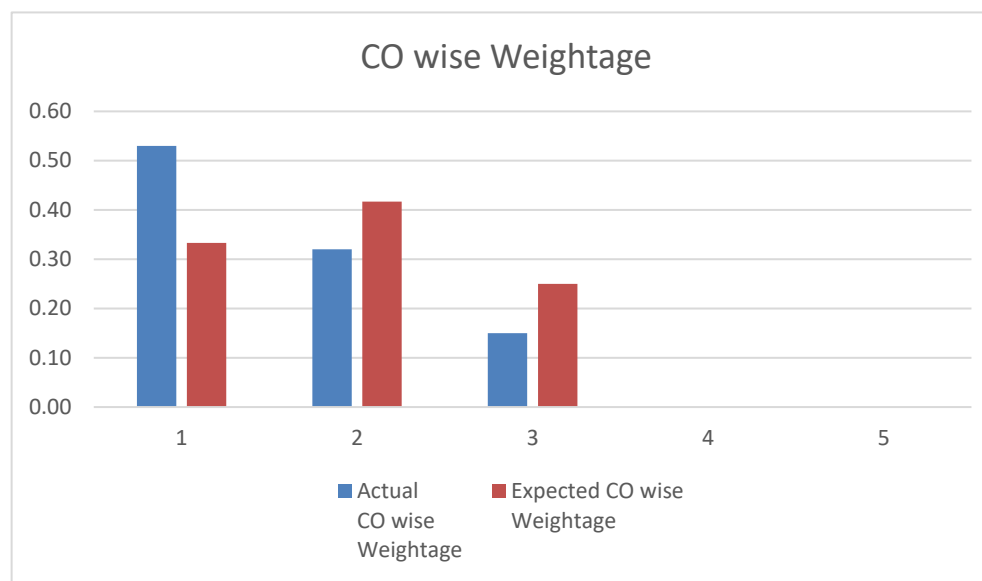
COs		Marks & Weightage				
CO No.	CO Name	MSE Marks (Max. 50)	Assignment Marks (Max. 20)	ESE Marks (Max. 100)	Actual CO wise Weightage	Expected CO wise Weightage
C01	Analyse the implementations of elementary data structures.	30	15	40	0.53	0.33
C02	Design and implementation of data structures like trees and graphs.	20	5	30	0.32	0.42
C03	Analyse the concepts of divide and conquer, dynamic programming, greedy, and back tracking techniques to efficiently solve problems.			30	0.15	0.25
	Marks (weightage)	0.3	0.2	0.5	1.00	1.00



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Note:

- In-semester Assessment is considered as the Internal Assessment (IA) in this course for 50 marks, which includes the performances in class participation, assignment work, class tests, MSE, quizzes etc.
- ESE for this course is conducted for a maximum of 100 and the same will be scaled down to 50 for grading.
- ESE marks for a maximum of 50 and IA marks for a maximum of 50 are added for a maximum of 100 marks to decide upon the grade in this course.

$$\text{Weightage for CO1} = (\text{MSE marks for CO1} / 1.6666 + \text{Assignment marks for CO1} / 1.0 + \text{ESE marks for CO1} / 2) / 100$$



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1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

Sl. No.	Tools	Weightage	Frequency	Details of Measurement (Weightage/Rubrics/Duration, etc.)
1	MSE	0.3	1	• Performance is measured using MSE attainment level. • Reference: question paper and answer scheme. • MSE is assessed for a maximum of 50 marks and scaled down to 30 marks.
2	Assignments	0.2	2	• Performance is measured using assignments/quiz attainment level. • Assignments/quiz are evaluated for a maximum of 20 marks.
3	ESE	0.5	1	• Performance is measured using ESE attainment level. • Reference: question paper and answer scheme. • ESE is assessed for a maximum of 100 marks and scaled down to 50 marks.

1.10 Course Articulation Matrix

CO	P01	P02	P03	P04	P05
C01					Y
C02				Y	
C03			Y		
Average Articulation Level			*	*	*